

Enhancing conflict detecting mechanism for Web Services composition: A business process flow model transformation approach

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Abstract

Businesses today are keenly aware of the competitive role of information technology. However, there is still a need for application systems to be implemented for support of daily business operations. However, these systems may be developed or acquired at different times from different vendors, and thus may be incompatible with each other. Because of this, a number of standards and technologies have been addressed to draw an integral vision in the field of enterprise application integration (EAI).

Emerging Web Services has now promised enterprises that they can deploy new digital services faster than ever before by integrating the existing application systems. By integrating in this fashion, operational processes of heterogeneous systems are seamlessly controlled and integrated through a meta-processes. This integration could preserve the autonomy of original systems, as well as enhance the flexibility and agility of the business. The researchers who constructed this study have attempted to explore in detail the design of these meta-processes. Also the researchers have proposed a conflict detecting mechanism (CDM) in order to aid process designers. A prototype with a case study has been developed to evaluate the feasibility of the CDM, and the results show that it is useful in guiding process designers to correct and improve their designed meta-processes.

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1. Introduction

When applying information technology (IT) for commercial purposes, businesses were gradually motivated by the progressive benefits and strategic advantages they could potentially realize. In the past, companies used to adopt or develop many different kinds of application systems to support their daily operations. However, as the degree to which a company relies on IT increases, the integration problems it will face as a result of IT reliance becomes

more complex [1]. The predominant problem occurring in many businesses today is that initial systems were developed or acquired at different times from different vendors, operated on different software or hardware technologies, and as a result are incompatible with each other [2]. This is the so-called “islands of information systems” phenomenon [3–5]. Because of the increased need for integration, standards, and technologies have been directed towards forming an integral vision in the field of enterprise application integration (EAI) [6].

The concept of using emerging Web Services has promised that enterprises can quickly deploy new digital services by integrating existing application systems [7]. When deploying this type of integration, the operational processes of heterogeneous systems are seamlessly controlled and integrated through meta-processes. This integration

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preserves the autonomy of original systems, as well as enhances flexibility and agility [8,9].

Organizations are now trying to move increase the process integration level due to greater expectations that it can yield large benefits. The achievement of this goal is still problematic due largely to that fact that it lacks a systematic approach to support the design of integral meta-processes [10–13]. Previous literature [11,14,15] had indicated that flaws in a designed business process would lead to high throughput times, low service levels, and a need for excess capacity. Therefore, in order to minimize risk, it is important to analyze the correctness, effectiveness, and efficiency of existing processes before they are changed [11,16]. Integral processes that contain errors may lead to frustrated customers, backlog, damage claims, and loss of goodwill.

Furthermore, there exists a gap between traditional process design and meta-process integration as enterprises migrate to Web-based services. Process designers are used to design business processes in the form of graphical diagrams, however, the meta-process designing approach requires representing different processes based on XML specifications.

As a result, in this proposed research, we define conflicts as any cause that could violate the normal or expected states of business process execution. In addition, we restrict the scope of this study to focus on conflict detection and design gaps of the meta-processes at the design stage when using Web-based services.

In this paper, we investigate the design of integral meta-processes and propose a conflict detecting mechanism (CDM), which plays an aiding role for process designers. We define conflicts as events that could lead to the designed integral meta-process violating the normal or expected results that it should be yielding. As a result, two main research questions are addressed in this paper and are explained in the following:

◆ Process design gap

There exists a gap between traditional process design and process integration when enterprises migrate to the use of Web-based services [17–19]. To aid in understanding existing processes, designers must represent business processes using an abstract graphical model. These integral meta-processes must be formatted in XML-based specifications. Next, Web-based Services composition engines could automatically execute the designed meta-processes. Unfortunately, writing XML-based process specifications only by a text-editor (which does not support useful XML character functions) is time consuming and erroneous work [13]. Not to mention the fact that it is difficult to ask traditional process designers to do such these jobs.

◆ Systematic designing guideline

Today, many organizations with sophisticated business processes have identified the need for concepts, techniques,

and tools to support the management of their business processes in an integrated manner. However, in the current market, the available business process management systems give limited support to systematically analyzing the possible conflicts of meta-processes during the design (or redesign) phase [11,12]. There needs to be a systematic approach to provide design guidelines that can support process designers, as well as allow them to correct and improve their designed meta-processes.

The remainder of this paper is organized as follows: In Section 2, we review related literature in the field of business process. In Sections 3 and 4, we present the design of solutions for two research problems. In Section 5, we describe the implementation of a prototype system, namely a CDM system. We evaluate our research utilizing a simulation and a real case study. The comparison between our research efforts and related works is provided in Section 6. The paper is concluded in Section 7.

2. Background information

2.1. Web Services composition

The trend of EAI has recently received considerable attention in the field of enterprise integration. However, the progress is still problematic as only a few EAI solutions are currently available. Lack of consensus in both the integration standards and the technologies is still a critical barrier to enterprises looking to adopt an EAI solution [1,5,13,20,21]. Among many proposed technologies, models, and architectures for EAI, an increased amount of attention has been given to a service-oriented architecture. This service-oriented architecture provides a vision to enterprises which allows them to leverage legacy systems into easily accessible, well described, and interoperable modular services [7,9]. Diverse application systems can be loosely coupled together through the implementation of these technologies [8,9].

For true integration of enterprise application systems, it is not sufficient to only implement the Web-based Services' standards. The most important step is to develop a management mechanism to integrate these enterprise services in an effective and efficient manner [8,22]. As a result, two specific terms, orchestration and choreography, appear to reflect the issue of Web Service composition [23].

When the underlying service-oriented architecture is implemented by Web Services, it becomes possible for companies to design, develop, and deploy a new enterprise service through meta-processes in a plug-and-play manner [13,24]. When BPEL describes different enterprise flows, it needs to acquire different services using UDDI. After acquiring these services by UDDI, it then needs to integrate with the BPEL flow. Therefore, UDDI is employed at the same level as BPEL. The vision proposed by Web Services composition and BPM is shown in Fig. 1.

Currently there are large amounts of research focused on the process composition of Web Services available. Tra-

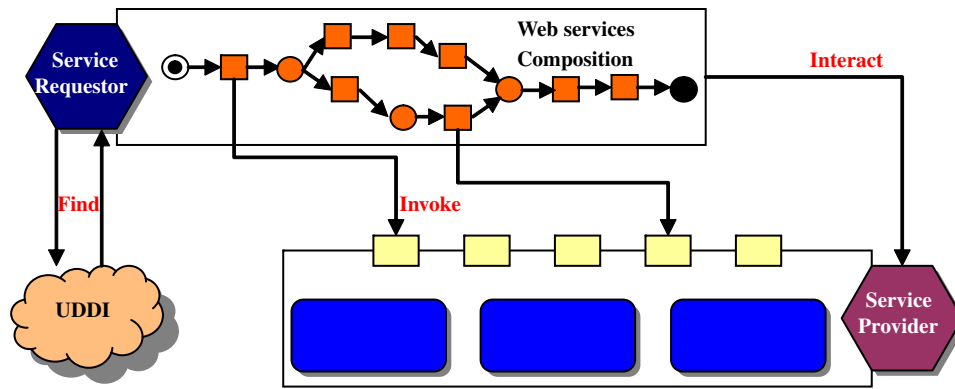


Fig. 1. The concept of Web Services composition.

ditionally, the conflict detection found in Web Services research can be classified as the following two types: type semantics mismatch [25–29] and policy mismatch [30,31]. In our research, we not only focus on semantics mismatch but also analyze possible conflicts when designers are participating in processes integration designs.

2.2. Business process modeling

Business process modeling has been the Holy Grail in the areas of IS development and workflow management. There were already over 10 kinds of process modeling methods proposed in the literature review and all are available in the marketplace. In order to increase understanding and aid in ease of use most of the modeling methods today adopt graphical diagrams to represent the abstract logic of business processes [11]. The most popular and widely known modeling techniques are Flowcharts [32], Petri Nets [33], Event-driven Process Chain (EPC) [34], UML Activity Diagram [35], Data Flow Diagram (DFD) [36], BPMN [18], and IDEF3 [37].

Since each different process modeling method has its own characteristics, the selection of an adequate method becomes an important issue for both academic researchers and business professionals. As a result, there are many research efforts dedicated to comparing [38–40] and transforming [41,42] these process modeling methods. In Table 1, we summarize the comparison of these major graphical process modeling methods according to the literature review.

In addition to graphical process modeling methods, another process modeling streams have received recent

attention. The emergence of XML has induced a paradigm shift in the software industry because of its powerful structural descriptive ability [43]. Thus, it is natural for standard established organizations to consider the possibility of representing business processes using XML. The major XML-based business process modeling specifications released are BPML [44], XPDL [45], BPEL4WS [46], WSCI [47], and BPSS [48]. Although different software vendors and organizations support different standards, there is still the need for debate in hopes of reaching a consensus amongst companies. In Table 2, we summarize a comparison of these major XML-based business process specifications according to the released standards and specifications [49,50]. In the field of Web Service composition, the predominant standard is the Business Process Execution Language for Web Services (BPEL4WS) which allows for the easy composition of Web Services but it does not utilize any user interaction. The model proposed by Lübke et al. provides the Web Service composition as well as the user interaction using EPC [51]. Furthermore, there are possibilities to map BPEL descriptions back to EPCs, according to other studies [52,53].

2.3. Business process analysis

Business process analysis is a critical step that must be approached in very carefully. Errors or flaws in the designed processes could result in unimaginable profit loss or system failure; especially when automation is utilized. Therefore, it is essential to detect possible conflicts in the design phase before the processes are put into production [11,54].

Table 1
The summary of graphical process modeling methods

	EPC	Petri Net	Flowchart	UML	DFD	BPMN	IDEF
Simplicity	High	Low	Very High	high	Low	Low	High
Power of notation	Very high	Very high	Medium	Medium	Very high	Very high	Medium
Model	Event based	Event based	Task based	Activity based	Functional based	Event based	Task based
Formalism	Some strength	Large strength	No	Passable	No	No	No
Standardizing	Yes	No	No	Yes	Yes	Yes	Yes
Analysis support	Large strength	Large strength	Passable	Some strength	Restricted/weak	No	Restricted/weak

Table 2
The summary of XML-based process specifications

	BPEL4WS	XPDL	BPML	WSCI	BPSS
Language	XML	XML	XML	XML	XML
Completeness	Very high	High	High	High	Medium
Support	Many strength	Restricted weak	Passable	Passable	Passable
Graphics	No	No	BPMN	No	UML
Products	Collaxa, IBM	No	Intalio	No	No
Standard	OASIS	WfMC	BPMI	W3C	ebXML
Extensible	High	High	High	High	Medium

Table 3
Summary of three major types of process analysis

Type	Target	Conflict examples
Validation	Syntactic conflicts	Illegal syntax or statements of business process specifications
Verification	Logical or semantic conflicts	Deadlocks or livelocks included in designed processes
Performance	Pragmatic conflicts	Process execution time violates organizational objectives

Traditional process analysis methods can be separated into three major types: validation, verification, and performance [11,16,54,55]. Validation analysis methods focus on discovering whether the designed business processes can be automatically enacted as expected. Through this analysis, possible syntax errors or violations should be detected and corrected [16,46,56].

Verification analysis methods are concerned with semantic or logical conflicts that are hidden in the designed processes and could lead to unsuccessful execution from start to finish. These kinds of analysis methods have been widely discussed in the literature review [14,41,56–59].

Finally, performance analysis methods focus on the quality of services (QoS) in the designed processes. There are also many research efforts [19,60,61] devoted to developing methods to analyze the performance of designed processes from different perspectives. In Table 3, we summarized these three major types of process analysis methods and the conflicts they are targeting.

3. Bridging process design gap

The first research question we addressed was the reason for the existence of a design gap between graphical process modeling and XML-based process specifications. As more enterprises want to embrace the benefits of Web Services composition, an effective transformation method should be developed to bridge such a gap. It should support process designers by directly transforming graphical process diagrams into the desired XML-based process specification like BPEL4WS or BPML.

Since own research focuses on the business process flow, the EPC diagram was chosen as the graphical presentation of the designed meta-processes. The EPC [62] and eEPC [63,64] are widely used due to their simplicity and can be employed to illustrate the workflow of the

information system as a chain of events and functions. One of the main reasons it has become so popular in practical business applications is because [41] the leading ERP vendor, SAP AG, uses it for representing R/3 processes, and it has developed many analytical techniques for business process reengineering (BPR). Also, a well-known market research organization, Gartner Group Inc., also places EPC as the worldwide leader for business process management.

Due to its high future potential, Web-based composition, proven research testing, and existing supported products, we adopted BPEL4WS as the XML-based process specification for describing meta-processes. In order to bridge such a design gap, the rules for transforming EPC into BPEL4WS should be developed to enhance automatic mapping.

First, we constructed four rules to transform the syntactical elements of EPC diagrams into BPEL4WS tags. These four rules are illustrated in the following:

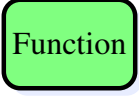
Rule 1 (Event). For every event diagram from the EPC, it can map the Receive, Invoke, or Reply constructs of BPEL4WS.

Example: A customer's order, Receive-Event in EPC can map to receive the tag of BPEL4WS. Formally as:

EPC Element	BPEL4WS Construct	Transformation
	<code><Receive></code> , <code><Reply></code> , and <code><Invoke></code>	<code><receive name="ABC" partnerLink="AService" portType="AportType" operation="initiate" variable="input"/></code>

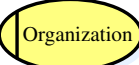
Rule 2 (Function). For every function diagram from the EPC, it can map to Assign the construct of BPEL4WS.

Example: A task responsible for copying the customer's order to handling service can map to the assigned tag of BPEL4WS. Demonstrated as:

EPC Element	BPEL4WS Construct	Transformation
	<code><Assign></code>	<pre> <assign> <copy> <from variable="input" part="parameters"/> <to variable="output" part="parameters"/> </copy> </assign> </pre>

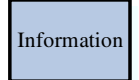
Rule 3 (Organization). For every organization diagram from EPC, it can map to the partnerLinks construct of BPEL4WS.

Example: An organization functioning as a service provider can provide different partnerLinks of services in the context of Web Services. Thus, an organizational diagram can map to the partnerLinks tag of BPEL4WS. Demonstrated as:

EPC Element	BPEL4WS Construct	Transformation
	<pre> <partnerLinks> <partnerLink> ... </partnerLinks> </pre>	<pre> <partnerLinks> <partnerLink name="XYZ" partnerLinkType="Xservice" partnerRole="XYZrole" myRole="ABCrole"/> </partnerLinks> </pre>

Rule 4 (Information.) For every information diagram from EPC, it can map to the variable construct of BPEL4WS.

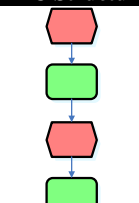
Example: Information is defined as required data or information necessary for task execution. Thus, an information diagram can map to the variables with a tag of BPEL4WS. Demonstrated as:

EPC Element	BPEL4WS Construct	Transformation
	<pre> <variables> <variable> ... </variables> </pre>	<pre> <variables> <variable name="XYZ" messageType="Xmessage"/> </variables> </pre>

In addition to syntactical elements transformation, we have constructed another four rules to transform the semantic structures of EPC into BPEL4WS tags. The details of these four rules are given in the following:

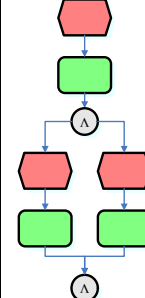
Rule 5 (Sequential pattern). For every sequential structure from EPC, it can map to the sequence construct of BPEL4WS.

Example: A list of activities linked in a sequential manner in EPC can map to the sequence tag of BPEL4WS. Demonstrated as:

EPC Structure	BPEL4WS Construct	Transformation
	<pre> <sequence> <activity1/> <activity2/> ... </sequence> </pre>	<pre> <sequence> <invoke name="XYZ" partnerLink="XService" portType="Y Service"/> <receive name="ABC" partnerLink="AService" portType="BportType"/> </sequence> </pre>

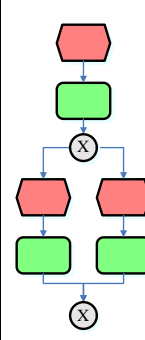
Rule 6 (Parallel pattern). For every parallel structure from EPC, it can map to the flow construct of BPEL4WS.

Example: An AND-Split structure of EPC contains several commands that should be executed simultaneously. Thus, an AND-Split structure can map to the flow tag of BPEL4WS. Demonstrated as:

EPC Structure	BPEL4WS Construct	Transformation
	<pre> <flow> <sequence1> <activity1/> ... </sequence1> <sequence2> <activity2/> ... </sequence2> </flow> </pre>	<pre> <flow> <sequence> <invoke name="XYZ" partnerLink="Xservice" portType="XportType"/> <receive name="ABC" partnerLink="Aservice"/> </sequence> <sequence> <invoke name="IJK" partnerLink="Iservice"/> <receive name="PQR" partnerLink="Pservice"/> </sequence> </flow> </pre>

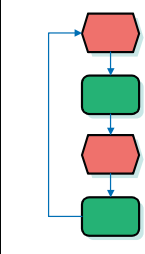
Rule 7 (Conditional pattern). For every exclusive choice structure from EPC, it can map to the flow construct of BPEL4WS.

Example: An X OR-Split structure of EPC containing two commands that should be executed exclusively. Thus, X OR-Split structure can map to the switch tag of BPEL4WS. Demonstrated as:

EPC Structure	BPEL4WS Construct	Transformation
	<pre> <switch> <case> <sequence> <activity/> ... </sequence> </case> <otherwise> <sequence> <activity/> ... </sequence> </otherwise> </switch> </pre>	<pre> <switch> <case condition="C1"> <assign> <copy> <from variable="v1"/> <to variable="v2"/> </copy> </assign> </case> <otherwise condition="C2"> <assign> <copy> <from variable="v3"/> <to variable="v4"/> </copy> </assign> </otherwise> </switch> </pre>

Rule 8 (Iterative pattern). For every loop structure from EPC, it can map to the while construct of BPEL4WS.

Example: Activities linked by an iterative structure in EPC can map to the “while” tag of BPEL4WS. Demonstrated as:

EPC Structure	BPEL4WS Construct	Transformation
	<pre> <while> <sequence> <activity> ... </sequence> </while> </pre>	<pre> <while condition="C"> <sequence> <invoke name="invoke" partner="IService" portType="IportType" operation="process"/> <assign> <copy> <from variable="v1"/> <to variable="v2"/> </copy> </assign> </sequence> </while> </pre>

In order to evaluate the accuracy of our transformation rules, we applied the method of information capacity [65]. This method has been used to guarantee that no information is lost while transforming between the two different models [66,67]. There are five levels of information capacity: Functional, Injective, Total, Surjective, and Bijection [65]. According to the study of Kwen and Fong, the transformation has to at least achieve an information capacity which preserves the mapping between the two models [66]. That is, the transformation rules satisfy the Total property as shown below:

- (I) Let A and B be different process model sets, where there exists a Functional relationship formally as $f: A \rightarrow B, \forall a \in A, \exists b \in B \cdot f(a) = b$.
- (II) Let S_1 be the original model, S_2 be the transformative model, and $I(S_n)$ represents the process structural set, there exists a Total relationship if and only if every structure of S_n satisfies the Functional property. Formally: $f: I(S_1) \rightarrow I(S_2), \forall I(S_1) \in S_1, \exists I(S_2) \in S_2$.

Accordingly, the structural transformation rules in our research, rules 5–8, all qualify as Total. That is, transforming the four basic structures from EPC to BPEL4WS demonstrates that information capacity preservation mapping has been successfully developed.

4. Systematic designing guideline

4.1. Conflict detecting mechanism (CDM)

The second research question addressed in this report concerns the methods required to support process designers in analyzing possible conflicts when they are participating in meta-process design. In the Web Services environment, the purpose of business process integration is to leverage existing enterprise application systems into new services. It is essential to detect possible conflicts contained in the integral processes before it is put into execution; this is a major factor in risk avoidance.

In order to achieve this goal, a systematic, step-by-step conflict detecting mechanism (CDM) is proposed in Fig. 2.

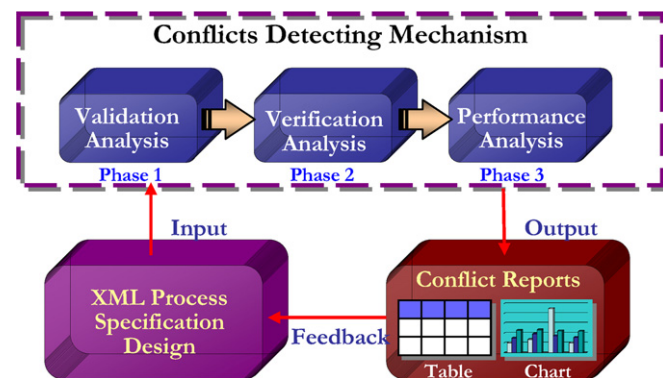


Fig. 2. Conflict detecting mechanism.

Unlike traditional analytical techniques that mainly focus on analyzing possible conflicts from process diagrams, the proposed CDM in this model focuses on the XML process specifications. The proposed CDM separates conflict detecting operations into a three-phased process analysis. The validation phase and verification phase (V&V) must be integrated in order to realize the improved quality of the process [68,69]. The ultimate goal of the validation and verification processes is to build the confidence that the software system is “fit for purpose”. This term refers to the system being able to handle its intended use. The third phase – performance analysis, is incorporated to check for pragmatic conflicts and also inspect for process execution times that violate organizational objectives. Once the integral XML process specification has been designed, the designer can then put the designed process into analysis. Three different process analysis phases are used to detect the three kinds of conflict as are discussed in this report.

During the first phase, designers check the process specifications for possible syntactical conflicts through validation analysis methods. The results of detected conflicts can give feedback to designers for further process redesign or debugging. Before moving forward to the next phase, it must achieve syntactical accuracy to guarantee executable property.

In the second phase designers invoke verification analysis methods to detect possible logical conflicts. These conflicts, if left unchecked, may lead to the unsuccessful execution of the process. Designers must remedy these conflicts in order to ensure that the designed process can be properly and repeatedly executed.

Finally, after finishing the previous two phases, the third phase is used to enhance the performance analysis methods that help designers detect possible pragmatic conflicts that violate organizational constraints. These results can help designers refine or select better-designed process alternatives. After finishing these phases, the CDM can again be utilized for the re-detection of possible conflicts for future process redesigns or debuggings.

4.2. Targeting conflicts

Before we implement CDM, it is essential to define what conflicts our research detected at each phase of CDM. The first barrier we faced was whether it would be possible to implement all process analysis methods mentioned in the literature review. The answer given time constraints was no. Thus, we applied the selection procedure proposed by Lawshe to justify the content validity of each conflict. The assessment procedure of content validity is used to ensure the accuracy or adequacy of the conflicts in our research [70].

To do this a steering committee was first formed to serve as the content evaluation panel. The committee was comprised of eight professionals from both industry and academic fields. Three committee members were from dif-

ferent universities, and five panelists were senior process designers currently in the business field. These eight committee members were asked to evaluate each conflict found in the literature on a three-point scale: “1= not relevant”; “2= important (but not essential)”; and “3=essential”. Based on the response, this study utilizes the following formula to calculate the content validity ratio (CVR) for each item:

$$\text{CVR} = (n - N/2)/(N/2)$$

where n is the frequency count of the number of committee members and N is the total number of panelists, namely $N = 8$ in this study.

The CVR for each conflict was determined for statistical significance at the 0.05 level. This study eliminates the conflicts for which CVR ratios are less than 0.75 [71]. Based on the outcomes of this procedure, we will illustrate details of each targeted conflict we implemented to guide designers in correcting and refining their designed meta-process.

To begin, we defined three conflicts that the validation analysis methods are targeted to detect. They are well formed, valid, and fit-for-composition conflicts:

- ◆ Well-formed conflict – because the BPEL4WS is an XML-based specification, it is the most important of all to guarantee that the syntax of the designed process obeys the well-formed property of XML [56].

Definition 1 (Well-formed). A designed BPEL4WS process specification is well formed and conflict-free if and only if:

- (I) For every designed BPEL4WS process (**BP**), it matches the production labeled document.
- (II) **BP** meets all the well-formed constraints released by the XML standard.
- (III) For all entities **PE** from **BP**, which are referenced directly or indirectly, still meet the well-formed property.

- ◆ Valid conflict – in addition to a well-formed conflict, validity is another essential property for the successful execution of BPEL4WS. Each designed process must be valid based on the data type declaration files to which it is referenced [46,56].

Definition 2 (Validity). A designed BPEL4WS process specification is valid and conflict-free if and only if:

- (I) For every designed BPEL4WS process (**BP**), it meets the well-formed property.
- (II) For every **BP**, it has an associated document type declaration in the form of XML schema.
- (III) For every XML schema (**XS**) associated with **BP**, it follows the released XML schema standard.
- (IV) For every **BP**, it complies with constraints expressed in the XML schema files it associated.

- ◆ Able-for-composing conflict – because BPEL4WS plays a composing role in existing services, it must follow the fit-for-composition guidelines. In our research, we named the third property as fit-for-composition, the goal of which is to guarantee the declarations referenced services and the required data are correct. They are represented by partnerLink and variable tags, respectively, in BPEL4WS specification [46,72].

Definition 3 (Fit-for-composition). A designed BPEL4WS process specification is fit-for-composition conflict-free if and only if:

- (I) For every designed BPEL4WS process **BP**, it contains referenced WSDL files **W**.
- (II) For every partnerLink and variable elements, **P** and **V** from **BP**, both are identical to the declarations in referred **W**. Formally: $\forall (BP \in P) \subseteq (W \in P)$ and $\forall (BP \in V) \subseteq (W \in V)$

In turn, we defined another three conflicts that the verification analysis methods are targeted to detect. They are starvation, lack of synchronization, and livelock conflicts:

- ◆ Starvation conflict – a starvation conflict would occur in the process specification whenever there has been a flaw in the structural design that leads to an X OR-Split instance ended with an AND-Join node [14,54], as depicted in the left part of Fig. 3.

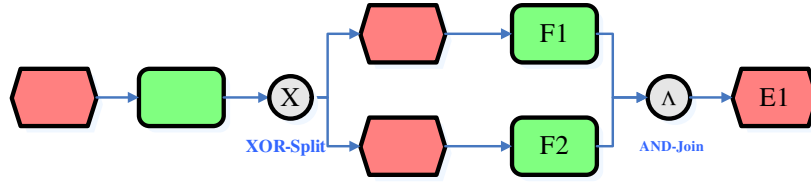
Definition 4 (Starvation). A designed BPEL4WS process specification contains the starvation conflict if:

- (I) For every Switch instance **S** from designed BPEL4WS process **BP**, there exists an AND-Join state **A** between two branches $\alpha 1$ and $\alpha 2$ from **S**. Formally: $\forall \text{BP}(\text{BP} \in S) \supseteq (S \in \alpha 1) \cap (S \in \alpha 2) = A$.

- ◆ Lack of synchronization conflict – the second conflict defined here is the lack of synchronization. It was strange that in some instances, the designed process had never been executed or had been executed twice. This is a result of another conflict in the structural design that draws an AND-Split instance with an OR-Join end node [14], as we can see in the center part of Fig. 4.

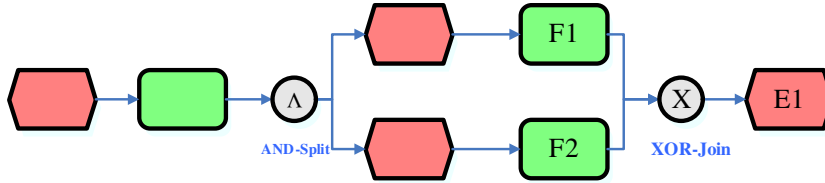
Definition 5 (Lack of synchronization). A designed BPEL4WS process specification contains the lack of synchronization conflict if and only if:

- (I) For every Flow instance **F** from designed BPEL4WS process **BP**, there exists an OR-Join state **O** among each branch βi from **F**. Formally: $\forall \text{BP}(\text{BP} \in F) \supseteq \cap (F \in \beta i) = O, i: \text{each branch}$.



Starvation for E1 can not synchronizing F1 and F2 in XOR-Split.

Fig. 3. Example of starvation conflict.



Lack of synchronous for F1 and F2 are ending with XOR-Join.

Fig. 4. Example of lack of synchronization conflict.

♦ **Livelock conflict** – without an appropriate end condition, the designed process which contains loop logic would fall into an infinite state and cause a livelock conflict. Therefore, the third type of conflict is livelock [46,54]. The right part of Fig. 5 gives an example of possible livelock situation.

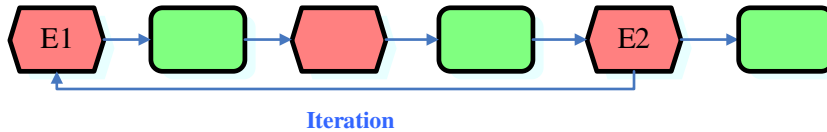
Definition 6 (Livelock). A designed BPEL4WS process specification contains the livelock conflict if and only if:

- (I) For every While instance ω from designed BPEL4WS process **BP**, there does not exist end condition μ from ω . Formally: $\forall \mathbf{BP}(\mathbf{BP} \in \omega) \supseteq (\omega \cap \mu) = \phi$.

- (II) For every expression variable v from μ , there exists an undeclared state U among v . Formally: $\forall \mathbf{BP}(\mathbf{BP} \in \omega) \supseteq (\mu \cap v) = U$.

Finally, we define a conflict that performance analysis methods are targeted to detect. It is the time-efficiency conflict:

- ♦ **Time-Efficient** – efficiency is an important indicator that an organization is concerned about completing their business in a timely and accurate manner. It is because of this that most commercial activities have certain tem-



Livelock for the loop does not have appropriate break condition at E2.

Fig. 5. Example of livelock conflict.

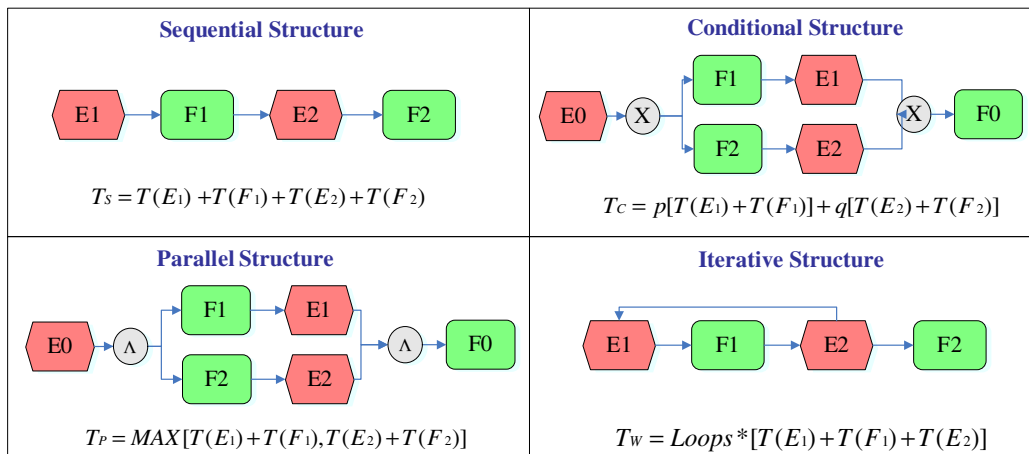


Fig. 6. Four circumstances for calculating time efficiency.

poral constraints [60,61]. For the sake of counting the total time efficiency of the designed process, we developed a calculating model according to four fundamental structures as shown in Fig. 6.

Definition 7 (Time efficiency). A designed BPEL4WS process specification contains the time-efficient conflict if and only if:

- (I) For every sequence instance activity A_i from the designed BPEL4WS process **BP**, the time performance is calculated formally by: $Ts = \sum_{i=1}^n A_i$.
- (II) For every Switch instance S_i from **BP**, every activity A_j from branch_1, every activity A_k from branch_2, and each branch execution probability α_i and β_i from S_i , the time performance is calculated formally by: $Tc = \sum_i \alpha_i (\sum_j A_j) + \beta_i (\sum_k A_k)$.
- (III) For every Flow instance F_i from **BP**, every branch B_j from F_i , and every activity A_{ijk} from B_j , the time performance is calculated formally by: $TP = \sum_i \max(\sum_j \sum_k A_{ijk})$.
- (IV) For every While instance W_i from **BP**, iterative times, the time performance is calculated formally by: $Tw = \lfloor f(T1, T2, T3) \rfloor$.
- (V) For every time performance $T = Ts + Tc + Tp + Tw$ from designed BPEL4WS process **BP** and every time performance constrain **C**, there exists a violated state δ . Formally: $\forall \mathbf{BP}(\mathbf{BP} \in T) \supseteq (T \cap C) = \delta$.

5. CDM system

5.1. System architecture

In order to support the procedures of the proposed CDM, a corresponding integrated developing environment (IDE) system, namely the CDM system, was developed. Fig. 7 demonstrates the system architecture of our CDM system, which aims to facilitate meta-process design and possible conflict detection.

The operation of our CDM system is based on a graphical user interface that interacts with the process designer at the first level. It also aids in bridging the process design

gap. Analysis methods for targeting conflicts are encapsulated into three conflict analyzers: validation, verification, and performance. It achieves the proposed CDM in a systematic manner. The third, repository layer, of the CDM system is to provide a warehouse function that can store the required data or information for the purpose of analysis. The primary implementation strategies of our CDM system are discussed in the following subsection.

5.2. System implementation

The implementation of our CDM system is based on the concept of a fully object-oriented integrated developing environment (IDE) that contains three fundamental class packages: diagram, specification, and analysis. These three packages are responsible for the graphical modeling requirements in the EPC, transforming the EPC into BPEL4WS, and the detection of possible conflicts based on BPEL4WS.

Every required class program must be encapsulated into these three packages according to its functional characteristics. By doing so, it is possible to exploit the advantages of an object-oriented model; these advantages being flexibility, reusability, and ease of replication [73]. Therefore, the function of the CDM system can be effectively achieved in a collaborative and integral fashion by implementing the objects of these three class packages.

As described previously, the diagram package must have enough diagram classes in order to fulfill the EPC modeling requirements of designers. After this is complete, the diagram objects can be put into the drawing toolbar, which provides designers with ease-of-modeling capabilities. Fig. 8 shows the UML class diagram of implemented diagram packages.

The second package needed is the specification package, which handles the transformation from EPC diagrams into BPEL4WS process specifications. The transformation rules we developed in the previous section for bridging the design gap are supported by specific class programs. Fig. 9 depicts the UML class diagram of the implemented specification package.

The third package, the analysis package, is the most important package in our CDM system. It supports the procedures of the proposed CDM and also encapsulates the three major analyzers in our CDM system. The analysis methods are implemented according to the targeted conflicts' definition as addressed above. Fig. 10 shows the UML activity diagram of the implemented analysis package.

6. Evaluation

6.1. Simulated integration scenario

In order to evaluate the feasibility of the proposed design gap bridging rules and the CDM, it is useful to set up a simulated run-time environment for Web Services

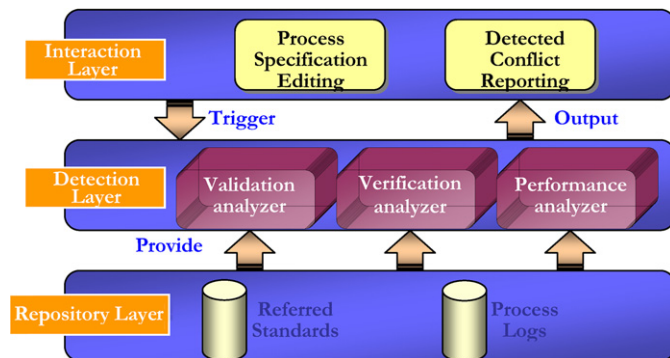


Fig. 7. Architecture of CDM system.

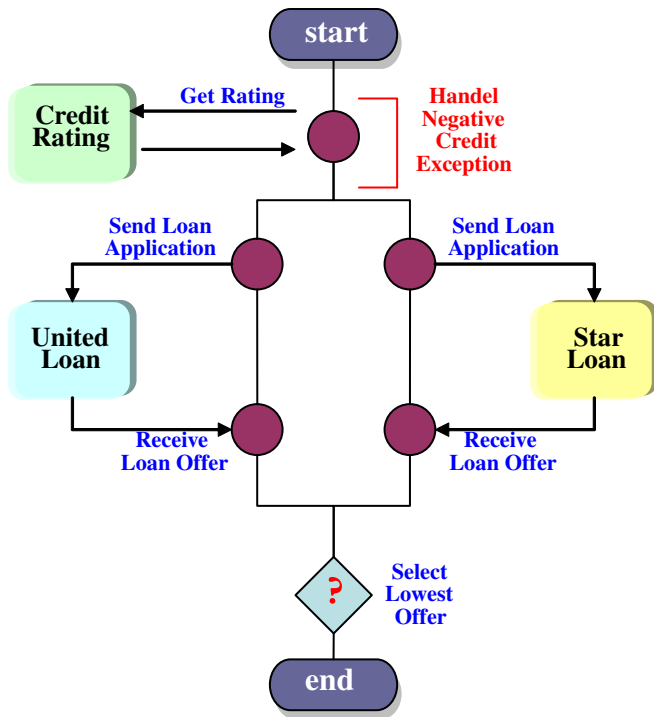


Fig. 11. LoanFlow example (source: Collaxa [74]).

then selects the most beneficial offer that is returned [74]. The LoanFlow scenario is a BPEL4WS example from the Collaxa BPEL Orchestration Server and can be downloaded directly from the Collaxa Website

According to Fig. 11, the LoanFlow example leverages the following services. A synchronized Credit Rating service which accepts a social security number as an input and returns an integer number as a credit rating. In addition, two non-synchronized loan processor services (UnitedLoan and StarLoan) are responsible for receiving loan application documents and returning loan offer documents. With the preparation of the testing environment and the LoanFlow scenario described above, we can now describe the implementation of our CDM system in a more specific and simple manner.

We first used our CDM system to design the graphical model of the LoanFlow meta-process in terms of EPC. Fig. 12 depicts the screenshot of our CDM system in supporting designers to draw the LoanFlow.

Next, we specified each element and attribute for generating the LoanFlow's BPEL4W specification via CDM system's GUI as shown in Fig. 13.

After finishing the specifying operations, the corresponding BPEL4WS specifications will be automatically generated as displayed in Fig. 14. These preparation steps also illustrate the feasibility of the implemented CDM system in bridging the process design gap.

As the designed meta-process was generated, we put it through the validation analysis to detect possible syntactical conflicts. The first phase of the CDM was built to aid designers in designing executable meta-processes. The validation analyzer in our CDM system is utilized to ensure the syntactical precision of the designed

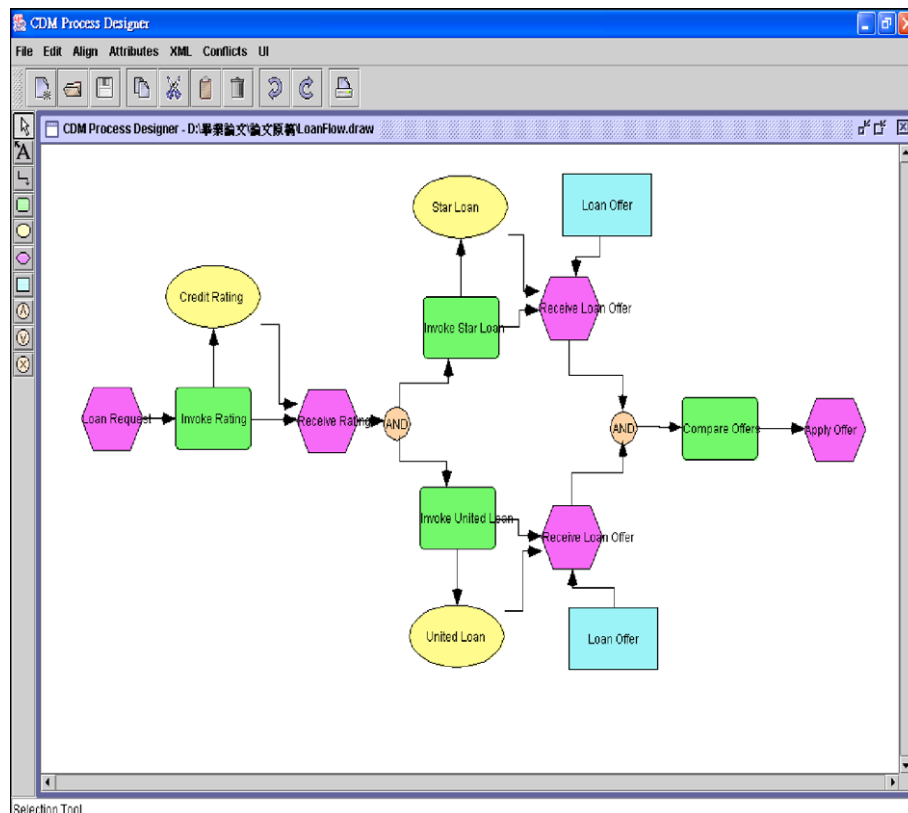


Fig. 12. LoanFlow meta-process design in EPC.

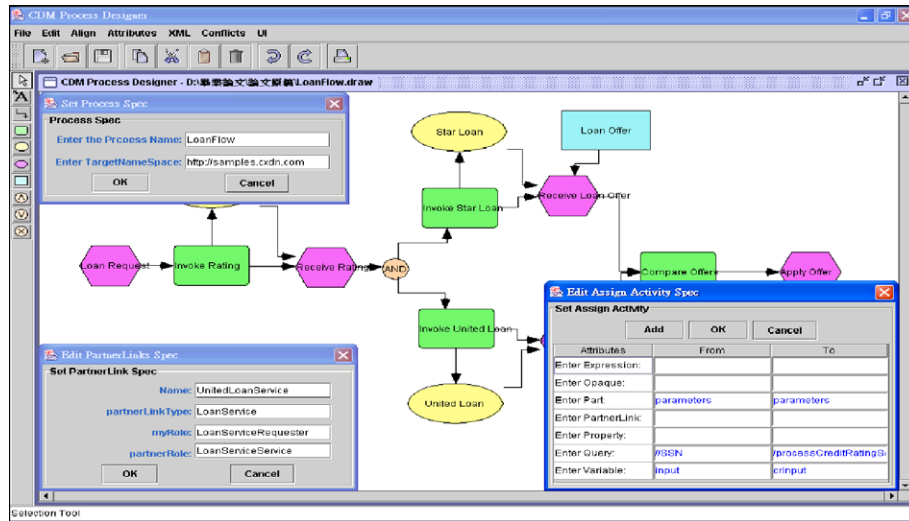


Fig. 13. Specify elements and attributes of LoanFlow.

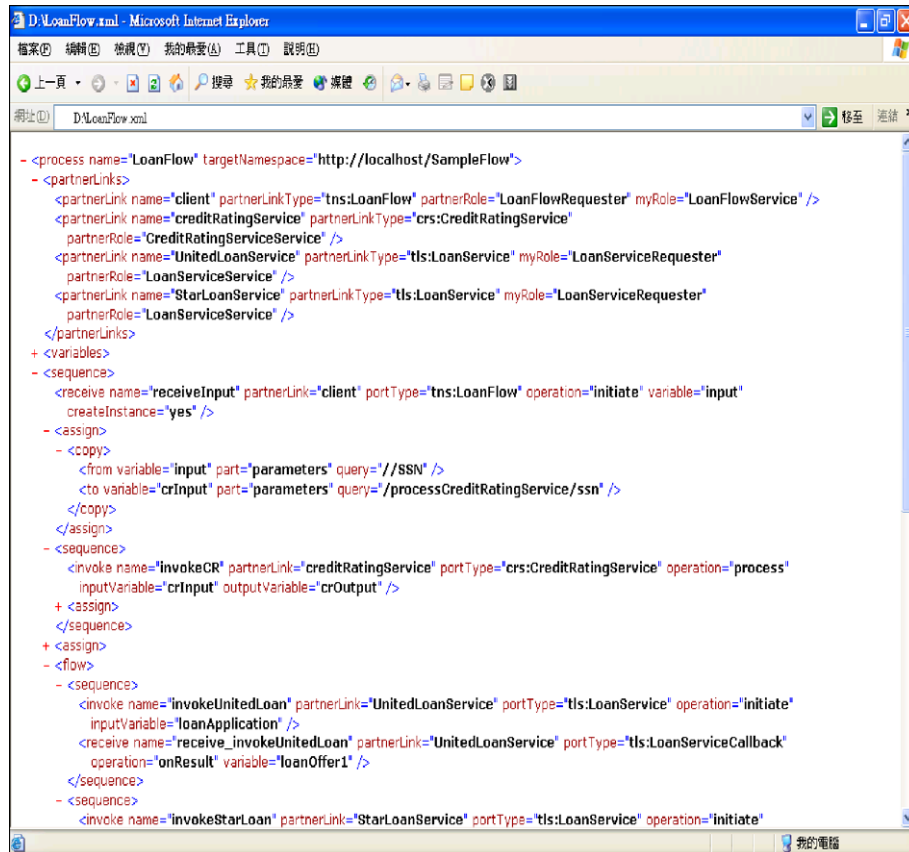


Fig. 14. Parts of generated LoanFlow's BPEL4WS specification.

BPEL4WS process specifications. It also provides different validation analysis methods in assisting designers to detect possible syntax violation situations in a BPEL4WS specification.

For testing purposes, the LoanFlow example is utilized to test the implemented validation analysis. The inputs of validation analysis are BPEL4WS files and the referenced

WSDL. The detected conflicts are represented in the form of a table, which effectively aids designers in completing the corrected operations. Process designers can easily locate the line of detected conflicts based on the analysis results and then quickly correct them. Fig. 15 is the screenshot of the detection results of the validation analysis of LoanFlow.

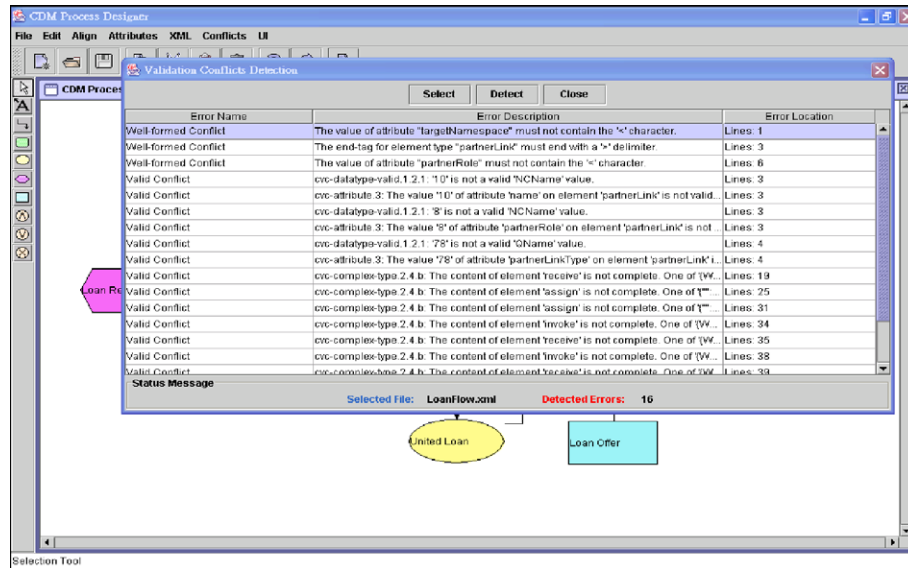


Fig. 15. Validation analysis results of LoanFlow.

The second phase of the CDM is a supportive role that assists designers in detecting possible structural or semantic conflicts. It provides three analysis methods which can be employed to detect starvation, lack of synchronization, and livelock conflicts as defined earlier in Section 4. All these analysis methods need to search the designed BPEL4WS specifications first to acquire the required data for analysis.

Process designers at this phase must design an integral meta-process that can be properly and repeatedly executed

from the start to the end of the activity. In turn, we put the LoanFlow into verification analysis and the detected conflicts are also represented in the form of a table as shown in Fig. 16.

The final analyzer in our CDM system is the performance analyzer. It is responsible for providing time performance analysis to detect possible time conflicts as defined earlier in Section 4. As long as the designed LoanFlow process operates successfully and continues operating as expected, it can then generate the historical

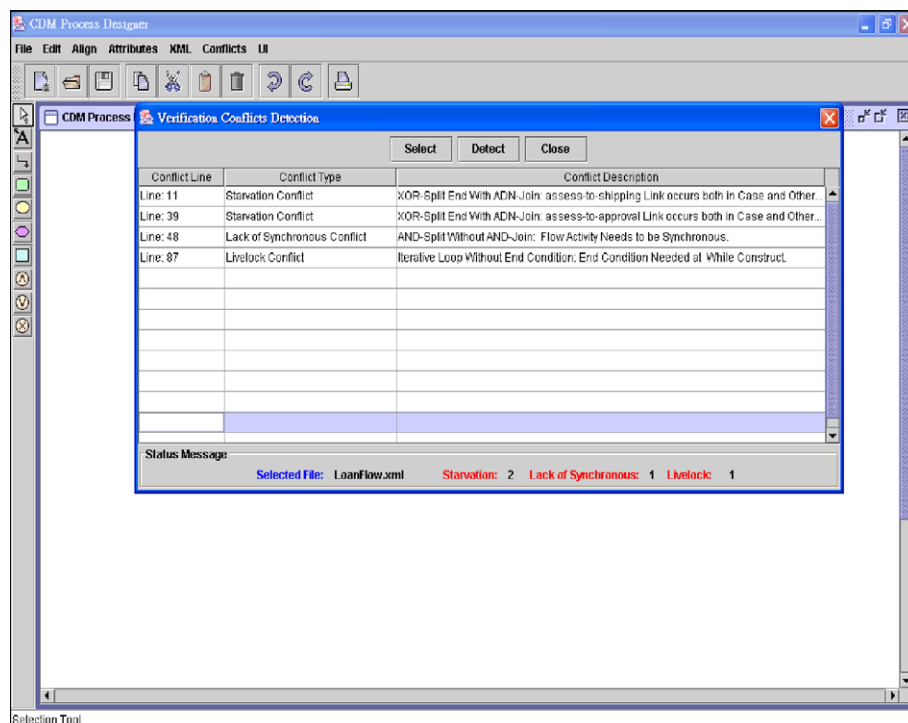


Fig. 16. Verification analysis results of LoanFlow.

execution time of each activity through repeated simulations. Thus, the average performance time of every activity can be input into performance analysis at the third phase of the CDM. First, we re-designed all three alternatives of the LoanFlow process. Next, we put these three process specifications into the Orchestration Server for execution. Each alternative process has been executed ten times to simulate historical records. The time logs were also stored in a database. Then, we set the time performance constraints as any average execution time of LoanFlow that exceeds 5 min. By doing so, we can put three specifications into the performance analyzer for possible time conflict detection. Different design alternatives of the LoanFlow can be compared from the perspective of time efficiency as demonstrated in Fig. 17. Process designers can then make the selection or refine the designed processes according to the analysis results.

Finally, with the implemented CDM system and the testing results, we can investigate the ordinal alternatives of the three phases in the proposed CDM. Let us set the validation analysis phase as α , verification analysis phase as β , and the performance analysis phase as γ . After testing, the other five alternatives could all possibly result in the discontinued situations as described below:

Alternative 1 ($\alpha \rightarrow \gamma \rightarrow \beta$). Set the order of detecting designed process conflicts to fail in this alternative if the designed process contains starvation conflicts before γ .

Alternative 2 ($\beta \rightarrow \alpha \rightarrow \gamma$). Set the order of detecting designed process conflicts in this alternative to fail if the designed process contains syntax conflicts before β .

Alternative 3 ($\beta \rightarrow \gamma \rightarrow \alpha$). Set the order of detecting designed process conflicts in this alternative to fail if the designed process contains syntax conflicts before β .

Alternative 4 ($\gamma \rightarrow \alpha \rightarrow \beta$). Set the order of detecting designed process conflicts in this alternative to fail if the designed process contains syntax conflicts before γ .

Alternative 5 ($\gamma \rightarrow \beta \rightarrow \alpha$). Set the order of detecting designed process conflicts in this alternative to fail if the designed process contains syntax conflicts before γ .

From the above discussion, only the order of $\alpha \rightarrow \beta \rightarrow \gamma$ in the proposed CDM is a better alternative to developing step-by-step conflict detecting guidelines.

6.2. Real integration scenario

In addition to the simulated LoanFlow scenario, we also applied a real integration scenario to evaluate the feasibility of our research works in practice. The case company is the Aerospace Industrial Development Corporation (AIDC) in Taiwan. It is a government-owned company with over 3000 employees and has annual revenues higher than NT\$ 10 billion. Traditionally, the major business scope of AIDC is to supply the domestic aerospace industry with design, manufacturing, testing, and maintenance services of aerospace-related products such as aircraft, engines, avionics equipment, and other parts. To meet the competitive demands of today's market, AIDC is now engaging in small-scale co-generation systems, cabin entertainment systems, medical care equipment, railways, and flight services.

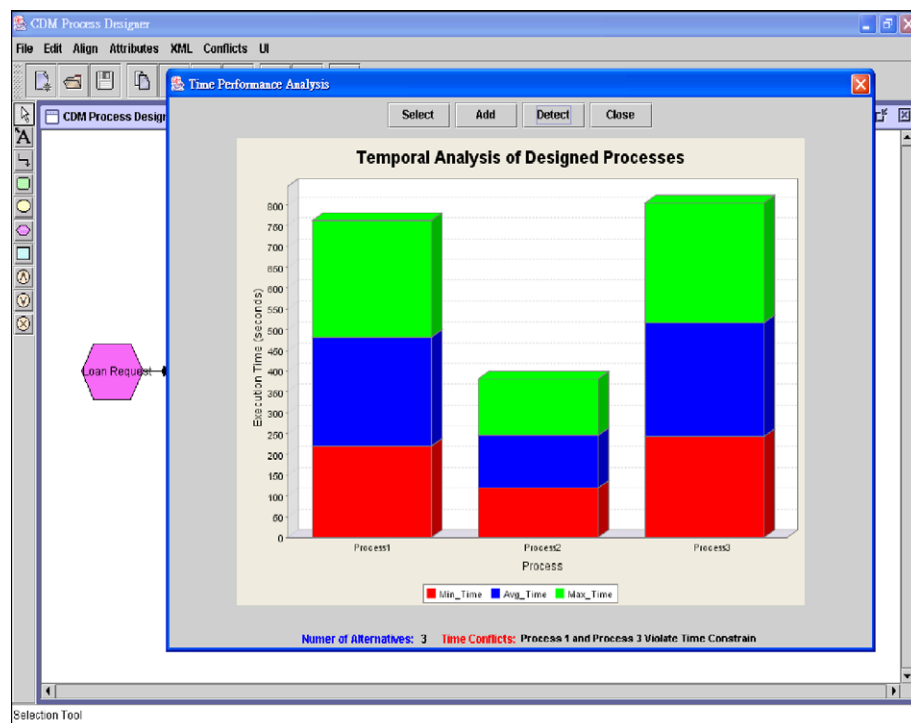


Fig. 17. Time efficiency analysis results of LoanFlow.

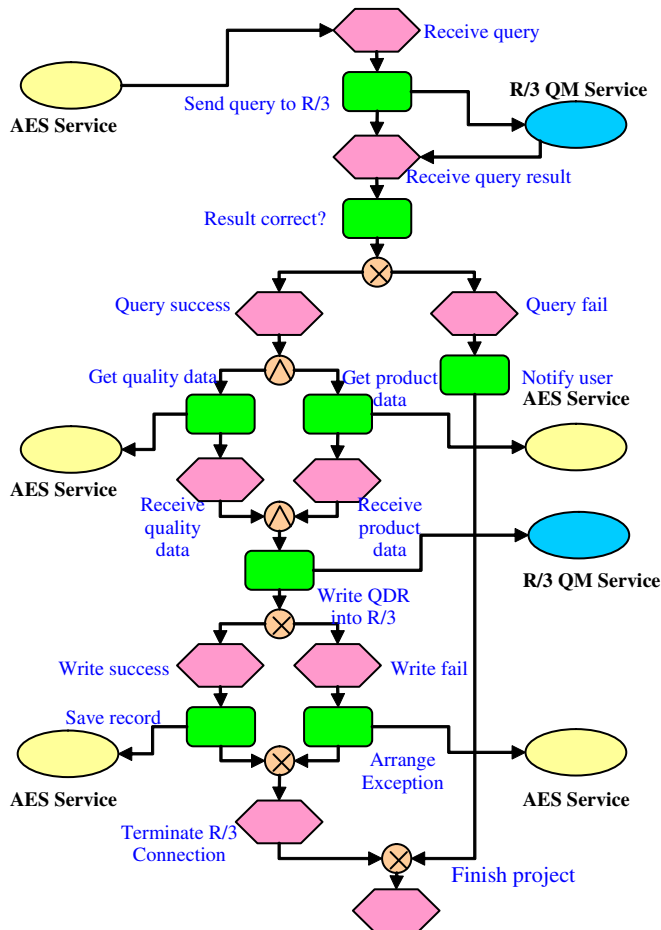


Fig. 18. Integral meta-process of QDR project.

The AIDC used to adopt large-scale packaged application systems, like Advanced Planning and Scheduling (APS) and Product Data Management (PDM) to support their business operations. In 1999, AIDC decided to implement the SAP R/3 ERP package for the sake of integrating all of the business activities in an integrated manner. As a result, the company is now facing integration problems caused by heterogeneous application systems.

For research purposes, we selected an on-going integration project, namely quality data records (QDR), which adopted the Web Services standards as a solution to integration problems. The objective of this project is to integrate the QDR process of in-house AIDC Engi-

neering System (AES), the R/3, with quality management module. The integral meta-process of QDR is depicted in Fig. 18.

To investigate the two research questions we used in the real integration case, we conducted an experiment with the project team. Ten QDR project team members were first instructed with the details of the proposed CDM and operations of the implemented CDM system. Each participant was then asked to design the integrated meta-process of the QDR under three circumstances. The first condition is to use the UltraEdit tool which can only support XML document editing. The second condition is to use the XML SPY tool which provides functions of graphical XML design and analyzation as well formed and valid properties of the designed specifications. The third condition is to use our CDM system which can support graphical meta-process design and enhancements for CDM provide guidelines to correct and refine the designed processes.

The experimental result is shown in Fig. 19. It indicated that with the functions of bridging process design gaps and CDM, process designers can design better integrated meta-processes. That is, the designed process contains less conflicts, especially in the area of syntactical conflicts. Our research also provided us with a positive assessment of perceived usefulness and end-user satisfaction from the subjects, which also implies the importance of our research.

6.3. A comparison with to related, prior researches

Recently, there are more and more related research efforts and commercial products that have begun being dedicated in the domain of Web Services composition and business process management. The major related works of our research are follows:

◆ Bind Studio v2.0

The Bind Studio v2.0 [75] is a commercial product developed by Bind Systems Ltd. which enables business analysts, process designers, and integration engineers to easily model business processes that comply with modern XML orchestration standards.

◆ BPEL Orchestration Server v2.0

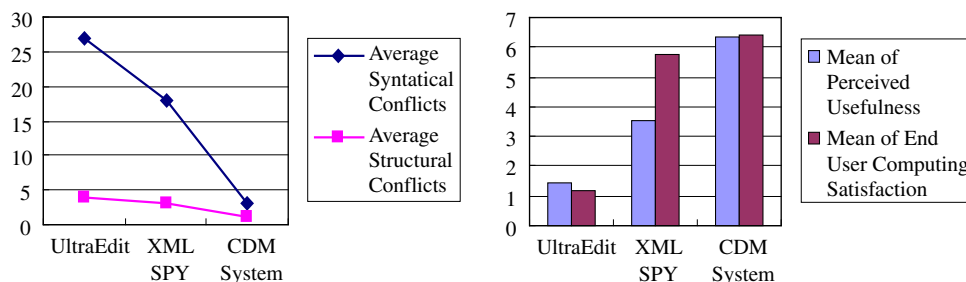


Fig. 19. Experiment results of real case study.

The BPEL Orchestration Server v2.0 [74] is the first commercial Web Services Orchestration Server released by Collaxa Inc. which enables the automation of business process integration via a pure-Java developed application server. It provides reporting, auditing, and debugging capabilities of BPEL specifications and makes it easy for any Java developer to publish synchronous and asynchronous Web Services and then, compose them into distributed process-centric applications.

◆ ebDesigner

The ebDesigner is a research effort addressed by Kim [76]. It aims to support dynamic setups of business processes among independent organizations through a formal standard schema. A formal editing tool called ebDesigner is used to first handle the task represented and managed B2B business processes using UML-compliant diagrams and then transforms the diagrams into an XML-based process specification.

◆ eFlow

The eFlow is the result of a research effort released by Fabio Casati et al. from HP Labs [77]. It supports the specification, enactment, and management of composite e-services. It uses modeled processes that are enacted by a service process engine. The objectives of eFlow are to develop composite e-services that can be employed to cope in the highly dynamic business environment and dynamically modify service process definitions in a simple and effective way to manage cases where user intervention is indeed required.

◆ SWORD

The SWORD is the research effort released by Shankar and Armando [78]. It is a toolset that allows service developers to quickly compose base Web Services to create new composite Web Services. It also works like a composition model which is well suited to typical informational Web Services. In addition, SWORD can work with today's Web Services unlike some other research/industry projects which are dependent on the deployment of standards.

◆ DAML-S

The DAML-S is a research effort by Narayanan and McIlraith (2002) [55]. It is an ontology for Web Services. The researchers exploit this to provide semantic markup of the content and capabilities of Web Services. It provides a graphical interface for process designers to input specifications based on DAML-S and provide functions of verification and simulation via the works of Petri Net.

◆ WCPT

The WCPT is a research effort by Chandrasekaran et al. [19]. It was developed for composing processes statically and storing them as Web Service Flow Language (WSFL) specifications. WPCT is capable of enacting the composed processes using its Perl enactment modules and simulating processes through different simulation models.

◆ SODA

SODA provides a Web-based design infrastructure that allows loosely coupled design teams to collaborate on different services, and to enable them to resolve any mismatch between heterogeneous design services. [79].

◆ WebDG

The WebDG system is implemented across a network of Solaris workstations. Citizens and case officers access WebDG via a graphical user interface (GUI) implemented using HTML/Servlet. WebDG currently includes seven FSSA applications implemented in Java (JDK 1.3). These applications are wrapped with WSDL descriptions. WebDG, provides customized services to indigent citizens. [80]

Bind System's Bind Studio v2.0 [75] and ebDesigner [76] were developed as a transformative method to automatically generate XML-based process specifications through UML activity diagrams. XRL/Flower [50] and DAML-S [55] on the other hand developed self-defined XML-based process specifications to transform XML into Petri Net. By doing so, the system embraces many of the analysis resources from the field of Petri Net modeling. Other investigations such as eFlow [77,81] and WCPT [19] have

Table 4
Summary of related works

Name	Graphical model	Specification	Validation	Verification	Performance	Target
Bind Studio	UML	BPEL4WS	Support	N/A	N/A	Process design gap
ebDesigner	UML	BPSS	N/A	N/A	N/A	Process design gap
eFlow	UML	Self-defined	N/A	N/A	Support	Process system
SWORD	Petri Net	Self-defined	N/A	Support	Support	Process design gap
DAML-S	Petri Net	DAML-S	N/A	Support	Support	IDE tool
WCPT	Flowchart	WSFL	N/A	N/A	Support	IDE tool
SODA	Design by authors	OWL-S	Support	Support	N/A	IDE tool
WebDG	Design by authors	WSFL	N/A	N/A	N/A	IDE tool
CDM (our work)	EPC	BPEL4WS	Support	Support	Support	IDE tool

adopted a traditional workflow model to design meta-processes and directly map to XML-based process specifications, as well as develop certain systems for further automation and simulation requirements. The related works are summarized in Table 4. From the above discussion, we can conclude that our research is unique because it provides a graphical interface for process designers to input process specifications based on BPEL4WS and functions of validation, verification, and performance analysis via the works of EPC. In addition, the CDM method can work with today's Web Services unlike some other emerging research/industry projects which simply rely on the deployment of standards.

7. Conclusion and limitation

In this report, the researchers addressed two fundamental questions in the field of meta-process integration. The results of this research indeed show that the process design gap and systematic design guidelines are important issues in supporting process designers throughout their works in the context of Web Services composition. We also found that there is an increasing number of related works focused on bridging the process design gap. However, to the researchers' current knowledge, there is still a lack research effort dedicated to systematically guiding process designers in step-by-step correction and refinement of designed meta-processes. There are still several issues that need to be addressed by future research projects, especially in regards to the complete transformation from EPC to BPEL4WS in a formal manner. Also an investigation of the impact of migrating traditional process analysis methods should be conducted.

Our research focuses on identifying process integration conflicts at design stage when enterprises are migrating toward Web Services. Consequently, the major contributions of our research include the following items.

- ◆ Providing a novel mechanism for guiding the detection of possible conflicts when designers are doing business process orchestration.
- ◆ Migrating the process analysis methods to the XML process specification through the implemented CDM system.
- ◆ Revealing the importance of bridging the gap between graphical business process modeling and the XML-base process specifications.
- ◆ Depicting the importance of conflict detection analysis at the process integration design phase via a real case study; and
- ◆ Providing a research foundation for future researchers who are interested in pursuing further studies in this area.

There are still several issues that needed to be addressed in future studies.

- ◆ Complete transformation from EPC to BPEL4WS in a formal manner

In our CDM system, we developed a straightforward transformation methodology to map basic EPC patterns to BPEL4WS. However, it may still need a complete and formal representative model to enhance inter-transformation of these two process models; it also needs a further guarantee that no information will be lost between either directions of the transformation.

- ◆ The impact of migrating traditional process analysis methods

In our research we implemented fundamental process analysis methods to enhance conflict detection. There are still many process analysis methods available in the domains of traditional workflow management and Petri Net. Future researchers can concentrate in investigating the impact of migrating process analysis methods to the XML process specifications.

- ◆ Interoperability among different process modeling methods; and

Currently, there have been more XML process specifications released. Our research adopted BPEL4WS for the representation of XML process specifications. However, soon companies could be facing the problem of interoperability among the released XML specifications.

- ◆ Efficiency of XML component breakdown;

The environment of Web Services relies heavily on XML-based standards. However, the work of breaking down an XML document is still efficient. Take our CDM system, for example, when the lines of the designed process specifications increase, it takes more time to do the analysis. Therefore, the improved analysis algorithm for XML dissection and conflict detection should be developed, especially in regards to a large business integration project.

References

- [1] J. Lee, K. Siau, S. Hong, Enterprise integration with ERP and EAI, *Communications of ACM* 46 (2) (2003) 54–60.
- [2] M. Themistocleous, Z. Irani, R.M. O'Keefe, R. Paul, ERP Problems and application integration issues: an empirical survey, in: *Proceedings of the 34th Annual Hawaii International Conference on System Sciences*, HICSS, 2001, pp. 3775–3784.
- [3] W. Hasselbring, Information system integration, *Communications of ACM* 43 (6) (2000) 32–38.
- [4] R. Chalmers, C. Campos, R. Grangel, References architectures for enterprise integration, *Journal of Systems and Software* 57 (3) (2001) 175–191.
- [5] N. Erasala, D.C. Yen, T.M. Rajkumar, Enterprise application integration in the electronic commerce World, *Computer Standards and Interfaces* 25 (2) (2003) 69–82.
- [6] D.S. Linthicum, *Enterprise Application Integration*, Addison-Wesley Longman, Reading, MA, 2000.
- [7] P. Fremantle, S. Weerawarana, R. Khalaf, Enterprise services, *Communications of ACM* 45 (10) (2002) 77–82.

- [8] F. Leymann, D. Roller, M.T. Schmidt, Web services and business process management, *IBM Systems Journal* 41 (2) (2002) 198–211.
- [9] H. Kregler, Fulfilling the web services promise, *Communications of ACM* 46 (6) (2003) 29–34.
- [10] J. Paul, P. Erik, Design principles for process modeling in enterprise application integration, *Information Systems* (26) (2001) 65–184.
- [11] A. Basu, A. Kumar, Research commentary: workflow management issues in e-business, *Information Systems Research* 13 (1) (2002) 1–14.
- [12] H. Smith, D. Neal, L. Ferrara, F. Hayden, The Emergence of Business Process Management, CSC Research Service Report, Available from: <http://www.cscresearchservices.com/process/>, 2002.
- [13] L. Charlesworth, T. Jones, The EAI and Web Services Report, *eAI Journal*, Available from: <http://www.eaijournal.com/>, 2003, 11–18.
- [14] W. Sadiq, M.E. Orlowska, Analyzing process models using graph reduction techniques, *Information Systems* 25 (2) (2000) 117–134.
- [15] W.M.P. van der Aalst, Loosely coupled interorganizational workflows: modeling and analyzing workflows crossing organizational boundaries, *Information and Management* 37 (2) (2000) 67–75.
- [16] A.P. Sheth, W.M.P. van der Aalst, I.B. Arpinar, Processes driving the networked economy, *IEEE Concurrency* 7 (3) (1999) 18–31.
- [17] H. Kim, Conceptual modeling and specification generation for B2B business processes based on ebXML, *ACM SIGMOD Record* 31 (1) (2002) 37–41.
- [18] S.A. White, Business Process Modeling Notation, BPMI.org, Draft 0.9, Available from: <http://www.bpmi.org/>, 2002.
- [19] S. Chandrasekaran, J. Miller, G. Silver, I. Arpinar, A. Sheth, Composition, performance analysis and simulation of web services, *Electronic Markets* 13 (2) (2003) 8–30.
- [20] W. van de Heuvel, W. Hasselbring, M. Papazoglou, Top-down enterprise application integration with reference models, *Australian Journal of Information Systems* 18 (1) (2000) 126–136.
- [21] C.C. Chiang, The use of adapters to support interoperability of components for reusability, *Information and Software Technology* 45 (3) (2003) 149–156.
- [22] W.M.P. van der Aalst, Don't go with the flow: web services composition standards exposed, *IEEE Intelligent Systems* 18 (1) (2003) 72–78.
- [23] C. Peltz, Web Services Orchestration: a Review of Emerging Technologies, Tools, and Standards, HP Technical Paper, Available from: http://devresource.hp.com/drc/topics/web_services.jsp/, 2003.
- [24] M. Stal, Web services: beyond component-based computing, *Communications of ACM* 45 (10) (2002) 71–76.
- [25] F. Casati, M.C. Shan, Event-based interaction management for composite e-services in eFlow, *Information Systems Frontiers* 4 (2) (2002).
- [26] R. Hamadi, B. Benatallah. A Petri net-based model for Web Service composition, in: K.-D. Schewe, X. Zhou (Eds.), *Database Technologies 2003 of the 14th Australian Database Conference (ADC 2003) Conference, Research and Practice in Information Technology*, vol. 17, Australian Computer Society, Adelaide, 2003, pp. 191–200.
- [27] B. Benatallah, M. Dumas, Q.Z. Sheng, Facilitating the rapid development and scalable orchestration of composite web services, *Distributed and Parallel Databases* 17 (1) (2005) 5–37.
- [28] L. Ardissono, L. Console, Anna Goy, Giovanna Petrone, Claudia Picardi, Marino Segnan, and Daniele Theseider Duppr'e, 'Cooperative model-based diagnosis of web services', in: *16th International Workshop on Principles of Diagnosis (DX 05)*, 2005, pp. 125–130.
- [29] K.P. Sycara, M. Paolucci, A. Ankolekar, N. Srinivasan, Automated discovery, interaction and composition of semantic web services, *Journal of Web Semantic* 1 (1) (2003) 27–46.
- [30] A. Rezgui, M. Ouzzani, A. Bouguettaya, B. Medjahed. Preserving privacy in web services, in: *Workshop on Information and Data Management*, 2002.
- [31] W. Xu, V.N. Venkatakrishnan, R. Sekar, I.V. Ramakrishnan, A framework for building privacy-conscious composite web services, in: *4th IEEE International Conference on Web Services (ICWS'06)*, Chicago, IL, September, 2006.
- [32] J.H. Harrington, E.K.C. Esseling, H. Van Nimwegen, *Business Process Improvement Workbook: Documentation, Analysis, Design, and Management of Business Process Improvement*, McGrawHill, New York, 1998.
- [33] K. Salimifard, M. Wright, Petri net-based modeling of workflow systems: an overview, *European Journal of Operational Research* 134 (3) (2001) 664–676.
- [34] A.W. Scheer, *Business Process Engineering – Reference Models for Industrial Enterprises*, second ed., Springer Verlag, Berlin/Heidelberg, 1998.
- [35] M. Dumas, M., A.H.M. ter Hofstede, UML Activity Diagrams as a workflow specialisation language, in: *Proceedings of the 4th International Conference on the Unified Modeling Language, UML, 2001*, pp. 76–90.
- [36] A.K. Penny, *Introduction to systems analysis & design: a structure approach*, IRWIN (1996).
- [37] R.J. Mayer, P.C. Benjamin, B.E. Carawaz, M.K. Painter, A Framework and a Suite of Methods for Business Process Reengineering, Available from: <http://www.idef.com/articles/framework/>, 1998.
- [38] V. Bosilj-Vuksic, V. Hlupic, Petri Nets and IDEF: applicability and efficacy for business process modeling, *Informatica* 25 (1) (2001) 123–133.
- [39] G. Mentzas, C. Halaris, S. Kavadias, Modeling business processes with workflow systems: an evaluation of alternative approaches, *International Journal of Information Management* 21 (2) (2001) 123–135.
- [40] E. Soderstrom, B. Andersson, P. Johannesson, E. Perjons, B. Wangler, Towards a framework for comparing process modeling languages, in: *Proceedings of the 14th International Conference on Advanced Information Systems Engineering, CAiSE02*, Toronto, Springer LNCS, 2002, pp. 600–611.
- [41] W.M.P. van der Aalst, Formalization and verification of event-driven process chains, *Information and Software Technology* 41 (10) (1999) 639–650.
- [42] A. Basu, R.W. Blanning, Workflow analysis using attributed metagraphs, in: *Proceedings of the 34th Annual Hawaii International Conference on System Sciences, HICSS, 2001*, pp. 3735–3743.
- [43] A.C. Lear, XML seen as integral to application integration, *IEEE IT Professional* 1 (5) (1999) 12–16.
- [44] A. Arkin, Business Process Modeling Language (BPML), BPMI.org, Draft.1.0, Available from: <http://www.bpmi.org/>, 2002.
- [45] R. Norin, Workflow Process Definition Interface–XML Process Definition Language, WfMC, 1.0 Final Draft, Available from: <http://www.wfmc.org/>, 2002.
- [46] T. Andrews, F. Curbera, H. Dholakia, Y. Golland, J. Klein, F. Leymann, K. Liu, D. Roller, D. Smith, S. Thatte, L. Trickovic, S. Weerawarana, Business Process Execution Language for Web Services Version 1.1, BEA, IBM, Microsoft, and SAP, Available from: <http://ifr.sap.com/bpel4ws/>, 2005.
- [47] A. Arkin, S. Askary, S., Fordin, W. Jekeli, K. Kawaguchi, D. Orchard, S. Pogliani, K. Riemer, S. Struble, P. Takacs-Nagy, I. Trickovic, S. Zimek, Web Service Choreography Interface 1.0", BEA, Intalio, SAP and Sun, Available from: <http://www.w3.org/TR/wsci/>, 2002.
- [48] J. Clark, C. Casanave, K. Kanaskie, B. Harvey, J. Clark, N. Smith, J. Yunker, K. Riemer, ebXML Business Process Specification Schema, ebXML, Available from: <http://www.ebxml.org/>, 2001.
- [49] R.K. Thiagarajan, A.K. Srivastava, A.K. Pujari, V.K. Bulusu, BPML: a process modeling language for dynamic business models, in: *Proceedings of Fourth IEEE International Workshop on Advanced Issues of E-Commerce and Web-Based Information Systems, WECWIS, 2002*, pp. 222–224.
- [50] W.M.P. van der Aalst, A. Kumar, XML-based schema definition for support of interorganizational workflow, *Information Systems Research* 14 (1) (2003) 23–46.

- [51] D. Lübke, T. Luecke, K. Schneider, J.M. Gómez, Using event-driven process chains for model-driven development of business applications, *XML4BPM* (2006) 265–280.
- [52] J. Mendling, J. Ziemann, EPK-Visualisierung von BPEL4WS Prozessdefinitionen, in: 7th Workshop Software-Reengineering (WSR2005), 2005.
- [53] J. Ziemann, J. Mendling, EPC-based modelling of BPEL processes: a pragmatic transformation approach, in: 7th International Conference on the Modern Information Technology in the Innovation Processes of the industrial enterprises MITIP, 2005.
- [54] W.M.P. van der Aalst, The application of Petri Nets to workflow management, *The Journal of Circuits, Systems and Computers* 8 (1) (1998) 21–66.
- [55] S. Narayanan, S. McIlraith, Simulation, verification and automated composition of web services, in: Proceedings of the 8th International World Wide Web Conference, WWW, Honolulu, Hawaii, Available from: <<http://www2002.org/CDROM/refereed/581/>>, 2002.
- [56] T. Bray, J. Paoli, E. Maler, Extensible Markup Language (XML) 1.0, second ed., W3C Recommendation, Available from: <<http://www.w3c.org/>>, 2000.
- [57] W. Reisig, G. Rozenberg, Lectures on Petri Nets I: Basic Models, LNCS, vol. 1491, Springer-Verlag, Berlin, 1998.
- [58] O. Marjanovic, M.E. Orlowska, On modeling and verification of temporal constraints in production workflows, *Knowledge and Information Systems* 1 (2) (1999) 157–192.
- [59] H. Zhuge, H.K. Pung, T.Y. Cheung, Timed workflow: concept, model, and method, in: Proceedings of the First International Conference on Web Information Systems Engineering, 2000, pp. 183–189.
- [60] L.J. Jin, F. Casati, M.C. Shan, Business Process Simulation with HP Process Manager, HP Technical Report, HPL-2001-285, Available from: <<http://www.hpl.hp.com/techreports/>>, 2001.
- [61] A. Kumar, W.M.P. Van der Aalst, H.M.W. Verbeek, Dynamic work distribution in workflow management systems: how to balance quality and performance, *Journal of Management Information Systems* 18 (3) (2002) 157–194.
- [62] Y. Yuan-Chen, C. Wei-Hao, L. Kun-Peng, Integration of EPC and a modularized colored Petri-Net through events for agile manufacturing cell control, in: Proceedings of the 2004 IEEE International Conference on Networking, Sensing & Control, Taipei, Taiwan, March, 2004, p. 9.
- [63] Y.C. Juan, C.O. Yang, Systematic approach for the gap analysis of business processes, *International Journal of Production Research* 42 (7) (2004) 9.
- [64] A.W. Scheer, ARIS – Business Process Modelling, third ed., Springer-Verlag, Berlin, 2000.
- [65] R.J. Miller, Y.E. Ioannidis, R. Ramakrishnan, The use of information capacity in schema integration and translation, in: Proceedings of the 19th International Conference on Very Large Data Base, VLDB, Dublin, Ireland, 1993, pp. 120–133.
- [66] I. Kwan, J. Fong, Schema integration methodology and its verification by use of information capacity, *Information Systems* 24 (5) (1999) 355–376.
- [67] S.M. Huang, H.Y. Hsueh, B. Lin, Information sourcing from internet and its verification, *Journal of Computer Information Systems* 42 (3) (2002) 94–105.
- [68] W.R. Adrion, M.A. Branstad, J.C. Cherniavsky, Validation, verification, and testing of computer software, *ACM Computing Surveys* 14 (2) (1982).
- [69] A.M. Davis, A.S. Zweig, The missing piece of software development, *Journal of Systems and Software* 53 (3) (2000) 205–206.
- [70] B.R. Lewis, C.A. Snyder, K.R. Rainer, An empirical assessment of the information resource management construct, *Journal of Management Information Systems* 12 (1) (1995) 199–223.
- [71] C.H. Lawshe, A quantitative approach to content validity, *Personnel Psychology* 28 (4) (1975) 563–575.
- [72] E. Christensen, F. Curbera, G. Meredith, S. Weerawarana, Web Services Description Language (WSDL) 1.1, W3C Note, Available from: <<http://www.w3.org/TR/wsdl/>>, 2001.
- [73] W. Sun, B.S. Lee, C.K. Yeo, JMS-Java-based meeting space, *Journal of Organizational Computing and Electronic Commerce* 10 (3) (2000) 171–188.
- [74] Collaxa Inc., BPEL Developer's Guide", Collaxa Technical Paper, Available from: <<http://www.collaxa.com/developer.docs.html/>>, 2003.
- [75] Bind Systems, Choreograph Web Services with BindStudio, Bind Systems White Paper, Available from: <<http://www.bindsys.com/>>, 2002.
- [76] H. Kim, Conceptual modeling and specification generation for B2B business processes based on ebXML, *ACM SIGMOD* 31 (1) (2002) 37–42.
- [77] F. Casati, S. Ilnicki, L. Jin, V. Krishnamoorthy, M.-C. Shan, eFlow: a Platform for developing and managing composite e-Services, in: Proceedings of AIWORC'2000, Buffalo, NY, April, 2000.
- [78] R.P. Shankar, F. Armando, SWORD: a developer toolkit for web service composition, in: Proceedings of the 11th International World Wide Web Conference, Honolulu, 2002, pp. 786–810.
- [79] M. Younas, K.-M. Chao, C. Laing, SODA: Service oriented design activities, in: Conference Proceedings of 8th International Conference on Computer Supported Cooperative Work in Design, IEEE Press, 26–28, May, 2004, Xiamen, China, pp. 419–424.
- [80] B. Medjahed, A. Bouguettaya, A.K. Elmagarmid, Composing web services on the semantic web, *VLDB J.* 12 (4) (2003) 333–351.
- [81] Fabio Casati, Ming-Chien Shan, Event-based interaction management for composite e-services in eFlow, *Information Systems Frontiers* 4 (2) (2002).